



# **StereoSplat+: Feed-Forward Gaussain Splatting with Diffusion-Assisted Progressive Inference**

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Meeting  
2026/05/26

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- Motivation & Current Problem

| Method                                              | Bin Average |      |        |       |
|-----------------------------------------------------|-------------|------|--------|-------|
|                                                     | RGB         |      | Depth  |       |
|                                                     | PSNR        | SSIM | AbsRel | SqRel |
| PixelSplat                                          | 19.04       | 0.60 | 0.094  | 0.579 |
| MVSplat                                             | 18.69       | 0.53 | 0.098  | 0.619 |
| OmniScene                                           | 18.19       | 0.53 | 0.117  | 0.657 |
| DepthSplat                                          | 18.60       | 0.60 | 0.140  | 0.767 |
| <b>StereoSplat<br/>(Ours)</b>                       | 20.42       | 0.67 | 0.071  | 0.386 |
| <b>StereoSplat+<br/>(bug: center gt leakage)</b>    | 21.03       | 0.73 | 0.067  | 0.360 |
| <b>StereoSplat+<br/>(revised with diffusion)</b>    | 19.89       | 0.66 | 0.074  | 0.422 |
| <b>StereoSplat+<br/>(revised without diffusion)</b> | 19.78       | 0.65 | 0.074  | 0.423 |

- StereoSplat+ injects enhanced pseudo views into StereoSplat.
- However, pseudo-view injection still degrades performance.
- Main reason: enhanced pseudo views are still distribution-shifted from GT views.

- **Possible Solutions to Improve the StereoSplat-plus over StereoSplat.**

**Path 1:** Train a stronger image enhancer to reduce the gap between rendered pseudo views and GT views. (better enhancer)

**Path 2:** Revise the StereoSplat training pipeline, for example with a two-stage strategy where some **GT views are replaced by pseudo-rendered views** in the second stage to improve robustness for StereoSplat+ inference. (robust base model)



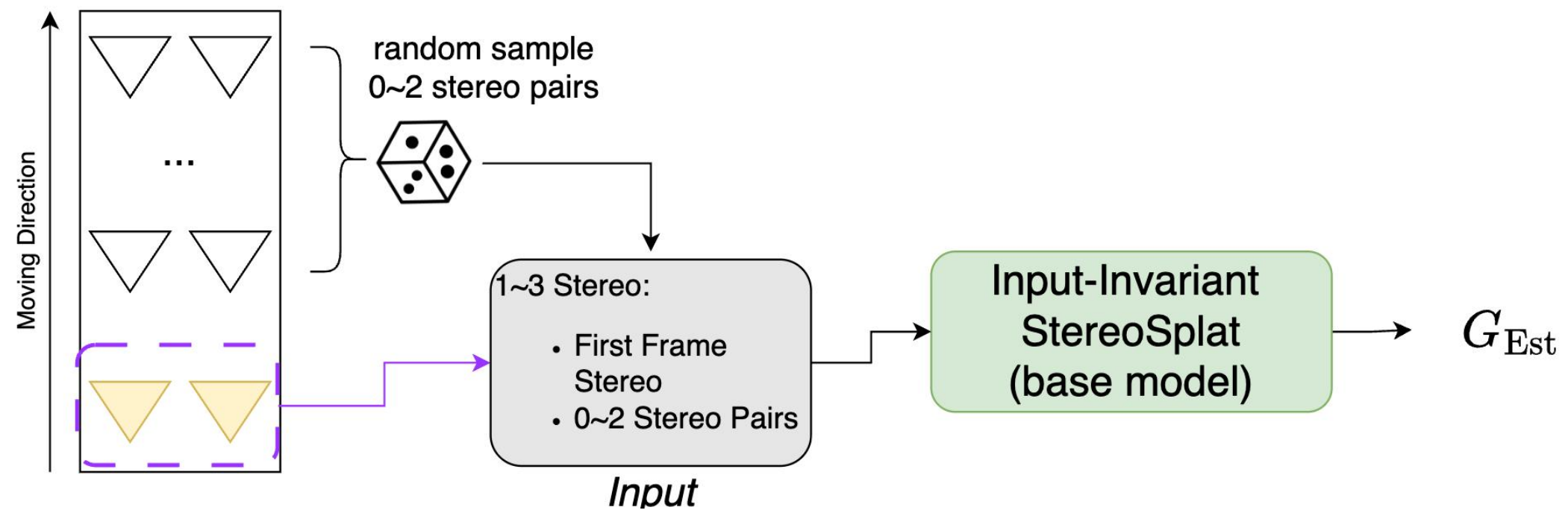
**Path 3:** Revise the inference stage by designing a more effective fusion strategy between  $G_{\text{base}}$  and  $G_{\text{plus}}$ . (robust inference manner)



This time, we focus on Path 2 and investigate whether pseudo-GT mixed training can improve the inference robustness of StereoSplat+.

- **Main Experiments Design of Path2**

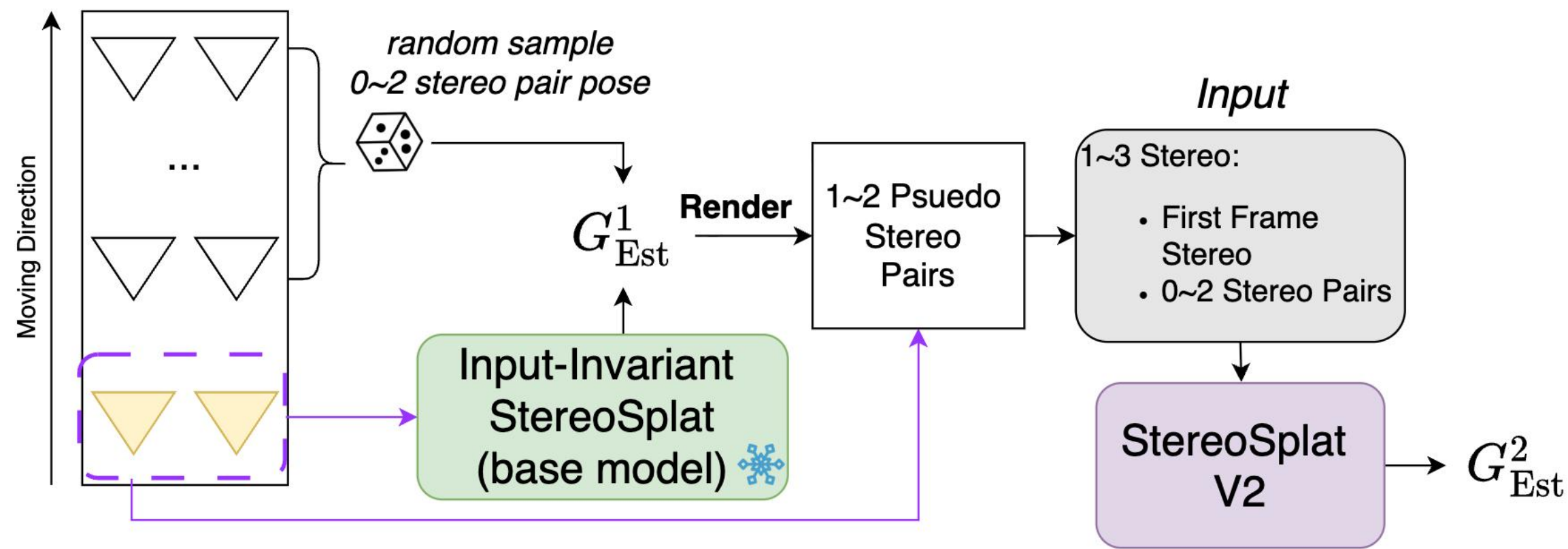
- Original Input-invariant StereoSplat Training



- Input: 1–3 stereo pairs, first-frame stereo pair is always included
    - Additional stereo pairs are randomly sampled from the bin
    - All input views are GT images

# • Main Experiments Design of Path2

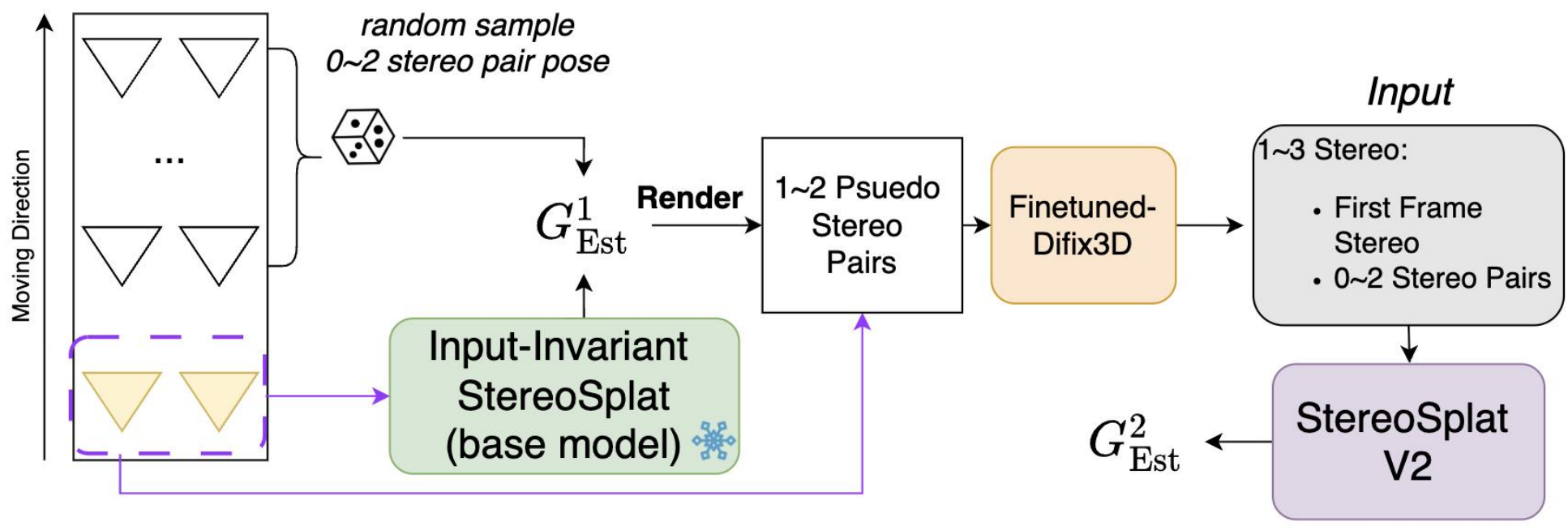
- Two-Stage Psuedo-GT Mixing Training **Last Meeting (05/12): No Difx3D**



- Train base model with the original setting GT views
- Use and frozen the base model to render pseudo views, Train V2 Model with mixed GT/pseudo inputs
- Additional sampled stereo pairs have 50% probability to be replaced by pseudo-rendered views

# • Main Experiments Design of Path2

- Two-Stage Psuedo-GT Mixing Training **Last Meeting (05/26): With Difix3D**



- Train the base model with the original GT-view setting, and freeze the base model to render pseudo views.
- Train the V2 model with mixed GT/pseudo inputs.
- Additional stereo pairs **are replaced by pseudo views with 50% probability.**
- For pseudo views, **50% are further enhanced by Difix3D.**

- # Evaluation Plan

Model Definition

**Model 1:** Stereosplat base model (train with GT)

**Model2:** Stereosplat-V2 (trained with pseudo-GT mix training, no difix3d enhanced)

**Model3:** StereoSplat-V3 (trained with pseudo-GT mix training, with difix3d enhanced)

**Exp1 : Normal StereoSplat 2 view input Setting**

- Model 1, first-frame stereo only
- Model 2, first-frame stereo only
- Model 3, first-fame stereo only

**Goal:** check whether Model's degrades under mix-clean input

**Exp. 2: StereoSplat+ inference with Model 2 pseudo views**

- Model 2 pseudo views + Difix3D
- Model 2 pseudo views without Difix3D
- Model 3 psuedo view with Difix3D
- Model 3 Psuedo view without Difix3D
- **Goal:** check robust model stereosplat for stereosplat-plus.

# Performance Evaluation

## Model Definition

- Model 1:** Stereosplat base model (train with GT)
- Model2:** Stereosplat-V2 (trained with pseudo-GT mix training, no difix3d enhaned)
- Model3:** StereoSplat-V3 (trained with pseudo-GT mix training, with difix3d enhanced)

## Exp1 : Normal StereoSplat Setting

| Method  | Input                           | Bin Average |      |        |       |
|---------|---------------------------------|-------------|------|--------|-------|
|         |                                 | RGB         |      | Depth  |       |
|         |                                 | PSNR        | SSIM | AbsRel | SqRel |
| Model 1 | First Frame Stereo<br>(2 Views) | 20.42       | 0.67 | 0.071  | 0.386 |
| Mode 2  | First Frame Stereo<br>(2 Views) | 20.45       | 0.67 | 0.073  | 0.383 |
| Model 3 | First Frame Stereo<br>(2 Views) | 20.65       | 0.67 | 0.067  | 0.324 |

It is quite superise that by using such kind of training, we discover that the model perform even better at the base settings improve from 20.42 to 20.66

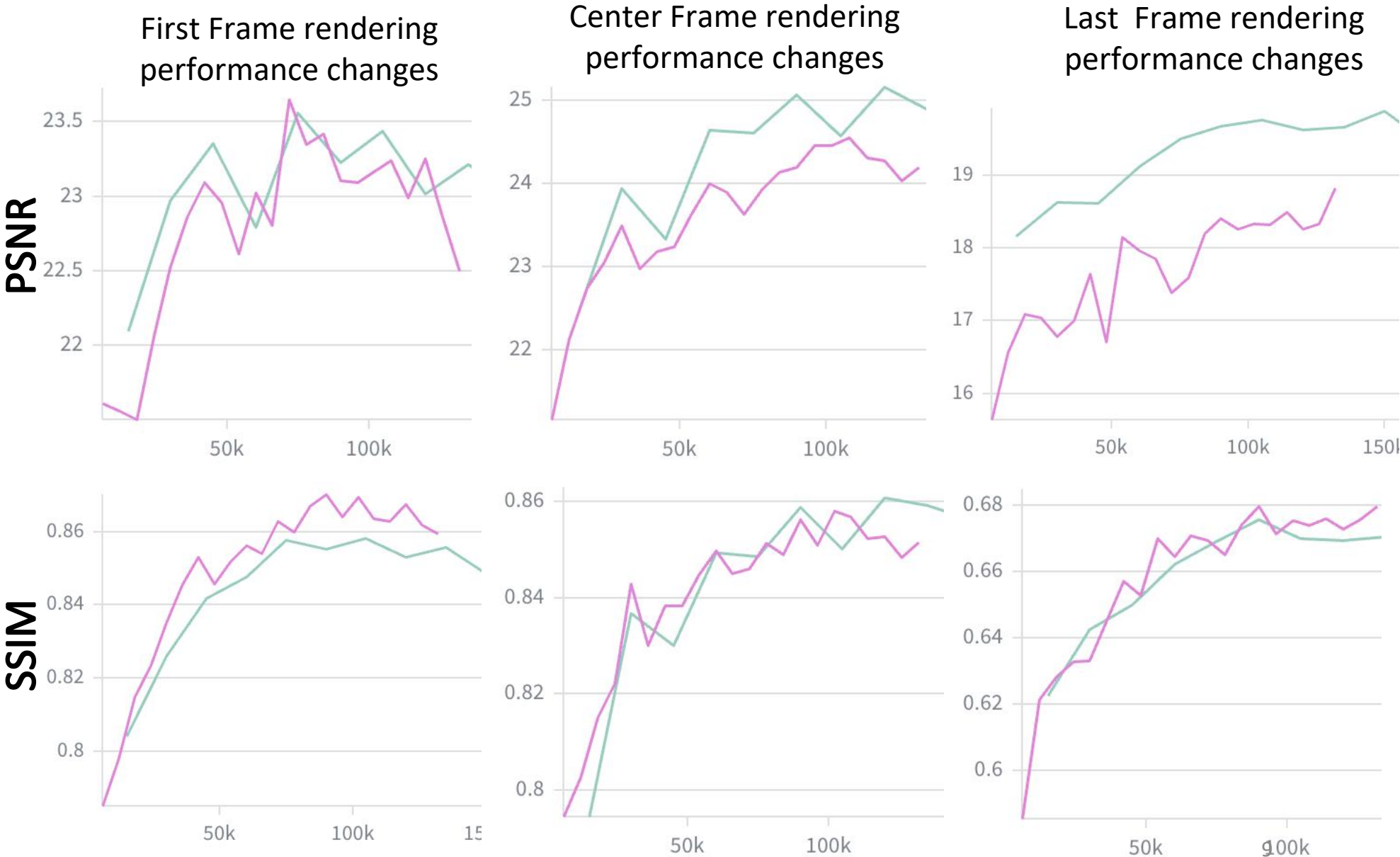


# Performance Evaluation

Model 2  
(mixed with no-  
difix psudeo)

Model 3  
(mixed with difix-  
enhance psudeo)

Difix3D improves PSNR,  
but not SSIM, likely due  
to geometry-  
inconsistent enhanced  
details.



# Performance Evaluation



Inside the red box are performance of the model trained with mix-training with difix3d enhancemnet.

| Method           | Input              | psuedo<br>view<br>Injected | Base Model | Use<br>Difix3D | Inference<br>Model | Bin Average |      |       |       |
|------------------|--------------------|----------------------------|------------|----------------|--------------------|-------------|------|-------|-------|
|                  |                    |                            |            |                |                    | RGB         |      | Depth |       |
|                  |                    |                            |            |                |                    | PSNR        | SSIM | MAE   | MSE   |
| StereoSplat      | First Frame Stereo | ×                          | ---        | ×              | Model 1            | 20.42       | 0.67 | 0.071 | 0.386 |
| StereoSplat      | First Frame Stereo | ×                          | ---        | ×              | Model 2            | 20.45       | 0.67 | 0.073 | 0.383 |
| StereoSplat      | First Frame Stereo | ×                          | ---        | ×              | Model 3            | 20.65       | 0.67 | 0.067 | 0.324 |
| StereoSplat-Plus | First Frame Stereo | √                          | Model 1    | √              | Model 1            | 19.89       | 0.66 | 0.074 | 0.422 |
|                  |                    |                            |            |                |                    |             |      |       |       |
| StereoSplat-Plus | First Frame Stereo | √                          | Model 1    | ×              | Model 2            | 20.34       | 0.67 | 0.074 | 0.391 |
| StereoSplat-Plus | First Frame Stereo | √                          | Model 1    | √              | Model 2            | 20.29       | 0.68 | 0.075 | 0.397 |
| StereoSplat-Plus | First Frame Stereo | √                          | Model 1    | ×              | Model 3            | 20.51       | 0.68 | 0.069 | 0.344 |
| StereoSplat-Plus | First Frame Stereo | √                          | Model 1    | √              | Model 3            | 20.56       | 0.69 | 0.069 | 0.334 |

# Performance Evaluation

| Method           | Input              | psuedo<br>view<br>Injected | Base Model | Use<br>Difix3D | Inference<br>Model | Bin Average |      |       |       |
|------------------|--------------------|----------------------------|------------|----------------|--------------------|-------------|------|-------|-------|
|                  |                    |                            |            |                |                    | RGB         |      | Depth |       |
|                  |                    |                            |            |                |                    | PSNR        | SSIM | MAE   | MSE   |
| StereoSplat      | First Frame Stereo | ×                          | ---        | ×              | Model 1            | 20.42       | 0.67 | 0.071 | 0.386 |
| StereoSplat      | First Frame Stereo | ×                          | ---        | ×              | Model 2            | 20.45       | 0.67 | 0.073 | 0.383 |
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Model 3 achieves the best overall performance and outperforms the GT-only Model 1 baseline even under the full StereoSplat+ pipeline.

However, the **progressive result (StereoSplat-Plus mode)** is still below the clean-input Model 3 baseline ( $20.56 < 20.65$ ), indicating that pseudo-view re-injection is still not fully beneficial.

# Performance Evaluation

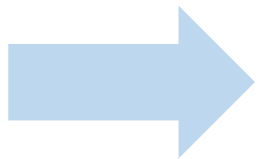
## Limitation of the Current Setting:

The current setting strictly follows the training input distribution **and inference by loading 2 kinds of models:**

- **Pseudo-view generation: Model 1**  
trained only with GT views
- **Final 3DGS re-generation: Model 2 / Model 3**  
trained with mixed pseudo-GT views, model 2 only using psuedo-GT mixed training, the psuedo view without using the difix3D enhancement. model 3 also using psuedo-GT mixed training, and the psuedo view use the difix3D enhancement.

This setting is effective because the inference distribution matches the training distribution of Model 2/3.

However, it has one practical limitation: **we need to load two different StereoSplat models during inference, which makes the pipeline less simple and less efficient.**



**How about using one model (Model 2 or Model 3) to do both psuedo-view generation and final 3DGS re-generation.(Idea solution)**

# Performance Evaluation

| Method           | Input              | psuedo<br>view<br>Injected | Base Model | Use<br>Difix3D | Inference<br>Model | Bin Average |      |       |       |
|------------------|--------------------|----------------------------|------------|----------------|--------------------|-------------|------|-------|-------|
|                  |                    |                            |            |                |                    | RGB         |      | Depth |       |
|                  |                    |                            |            |                |                    | PSNR        | SSIM | MAE   | MSE   |
| StereoSplat      | First Frame Stereo | ×                          | ---        | ×              | Model 1            | 20.42       | 0.67 | 0.071 | 0.386 |
| StereoSplat      | First Frame Stereo | ×                          | ---        | ×              | Model 3            | 20.65       | 0.67 | 0.067 | 0.324 |
| StereoSplat-Plus | First Frame Stereo | √                          | Model 1    | √              | Model 1            | 19.89       | 0.66 | 0.074 | 0.422 |
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| StereoSplat-Plus | First Frame Stereo | √                          | Model 1    | ×              | Model 3            | 20.51       | 0.68 | 0.069 | 0.344 |
| StereoSplat-Plus | First Frame Stereo | √                          | Model 1    | √              | Model 3            | 20.56       | 0.69 | 0.069 | 0.334 |
| StereoSplat-Plus | First Frame Stereo | √                          | Model 3    | ×              | Model 3            | 20.41       | 0.57 | 0.070 | 0.356 |
| StereoSplat-Plus | First Frame Stereo | √                          | Model 3    | √              | Model 3            | 20.47       | 0.68 | 0.069 | 0.336 |

As shown in the results, using only one model for inference leads to worse performance. This is likely because the inference setting becomes inconsistent with the training input distribution.

# Current Conclusion